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COURSE CODE: SBT-DF202

LAB NAME: LAB3-DATA CARVING WITH XXD, BINWALK and SCALPEL

DATE: 05/09/26

INSTRUCTOR: AMINU IDRIS

SCENARIO

Your practical context

A forensic storage image contains files that may no longer be recoverable through normal file-system browsing. Some file metadata may be missing, damaged, or unreliable. Acting as a junior digital forensic analyst, you are required to examine raw file content, identify file signatures, recover embedded content, and carve deleted files from a USB forensic image while documenting every step in a repeatable forensic manner.

OBJECTIVES

What you must demonstrate

By completing this lab, students will demonstrate their ability to explain why data carving is required when file-system metadata is missing or unreliable; use xxd to examine binary file content and identify headers and footers; recognise JPEG SOI and EOI signatures; reconstruct binary content from a hexadecimal dump; use The Sleuth Kit to examine inode, block and sector-level data; identify and extract embedded content using Binwalk; recover deleted files from a forensic USB image using Scalpel; verify recovered evidence using cryptographic hashes; and prepare a professional forensic report containing commands, screenshots, findings and limitations.

INSTRUCTOR BRIEFING

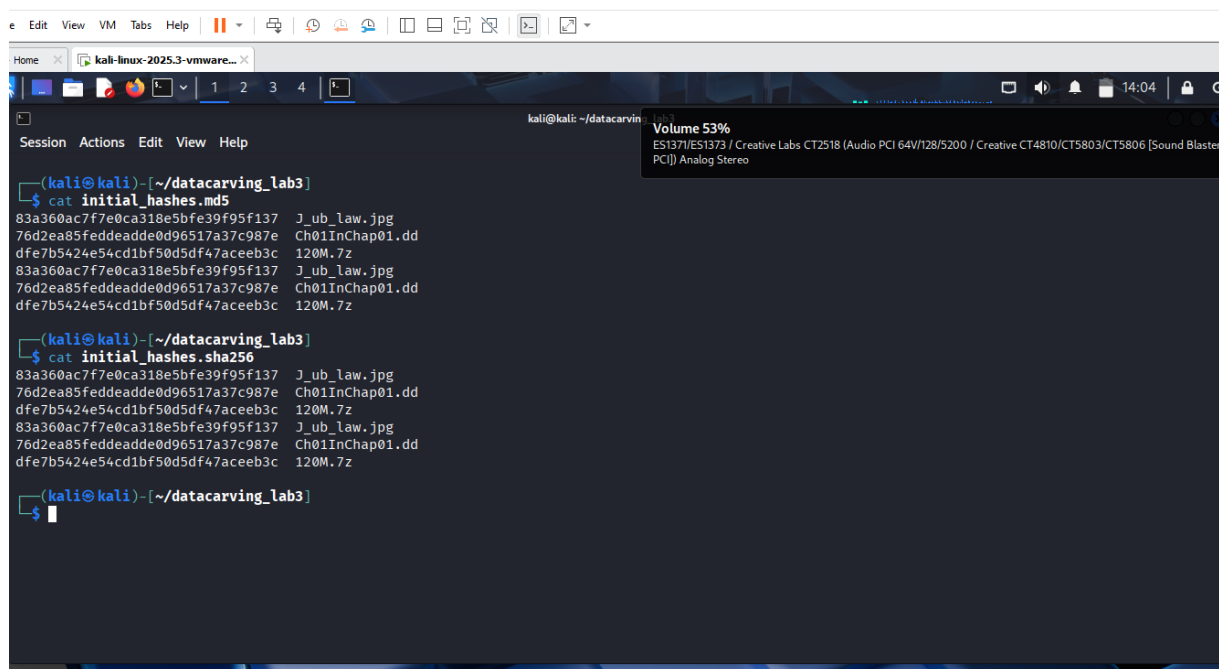
Laboratory guidance

Before beginning this lab, review the **Module 7 lessons and study material on Data Carving and File Recovery Techniques**. The lecture material is already a detailed walkthrough, so you

are expected to understand file signatures, hexadecimal analysis, embedded-file recovery and carving concepts before executing commands.

Create a dedicated working directory and preserve all downloaded evidence files. Record MD5 and SHA-256 hashes before beginning analysis.

ANSWER:



```
kali@kali: ~/datacarving_lab3
$ cat initial_hashes.md5
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
76d2ea85feddeade0d96517a37c987e Ch01InChap01.dd
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
76d2ea85feddeade0d96517a37c987e Ch01InChap01.dd
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z

kali@kali: ~/datacarving_lab3
$ cat initial_hashes.sha256
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
76d2ea85feddeade0d96517a37c987e Ch01InChap01.dd
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
76d2ea85feddeade0d96517a37c987e Ch01InChap01.dd
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z

kali@kali: ~/datacarving_lab3
$
```

Required Lab Resources

Download the authorised training evidence below:

- [Ch01InChap01.dd — Forensic DD Image](#)
- [J_ub_law.jpg — JPEG Sample for Hex and Signature Analysis](#)
- [120M.7z — USB Image for Scalpel File Carving](#)

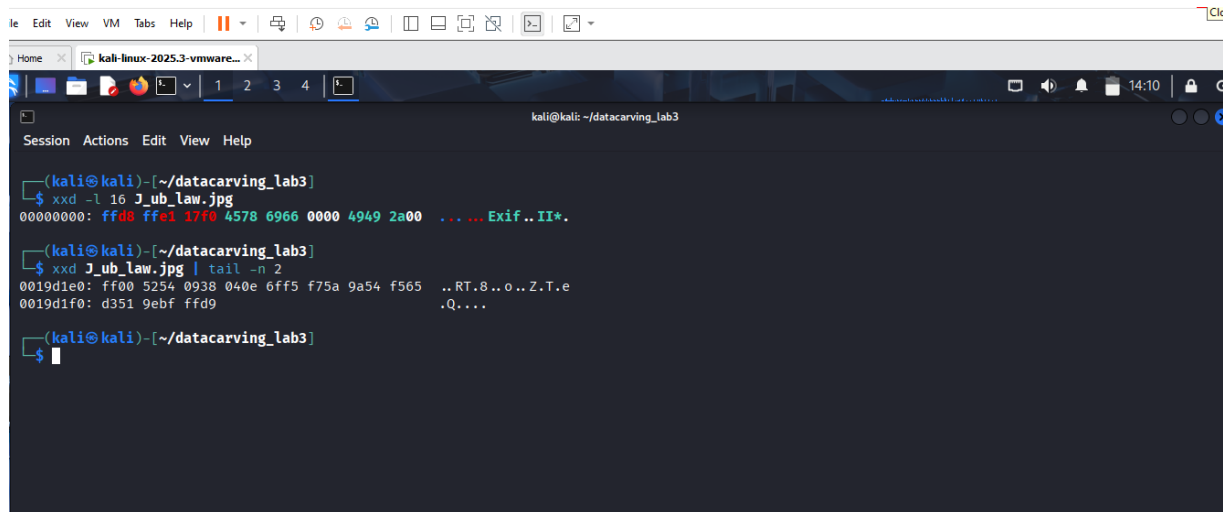
These are the exact evidence resources identified in the lab guide.

Part A - File Signatures and Hex Analysis

Use xxd to inspect J_ub_law.jpg, identify the JPEG header and footer signatures, create a plain hexadecimal dump, and reconstruct the image using reverse-hexdump functionality.

Verify the reconstructed image by comparing MD5 and SHA-256 hashes with the original.

ANSWER:

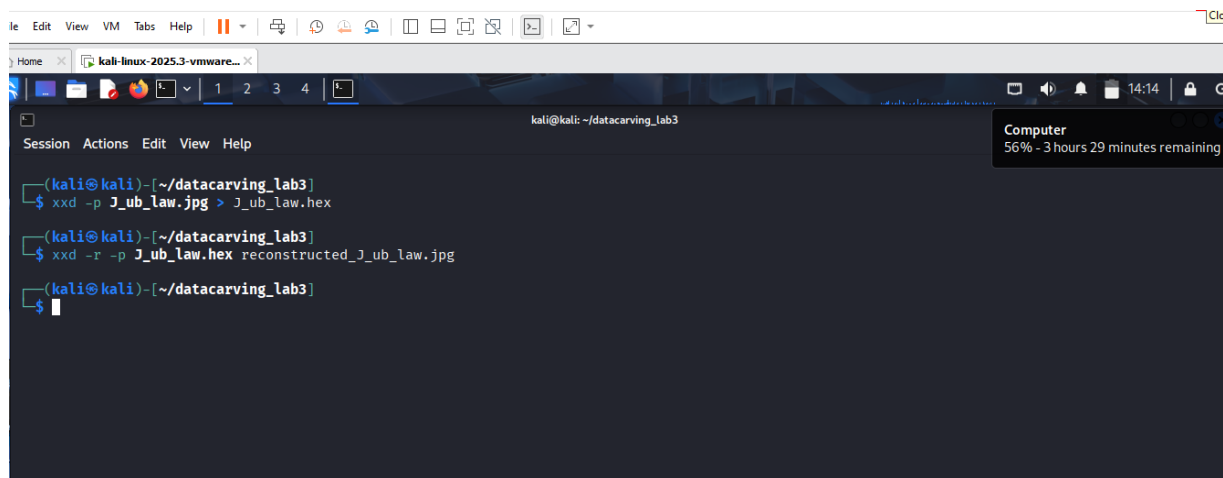


A terminal window titled 'kali@kali: ~/datacarving_lab3' showing the execution of two xxd commands. The first command, 'xxd -l 16 J_ub_law.jpg', displays the first 16 bytes of the file in hexadecimal and ASCII, identifying the 'Exif..' header. The second command, 'xxd J_ub_law.jpg | tail -n 2', displays the last two lines of the hex dump, showing the end of the file data.

```
(kali@kali)-[~/datacarving_lab3]
$ xxd -l 16 J_ub_law.jpg
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ... Exif.. II*.

(kali@kali)-[~/datacarving_lab3]
$ xxd J_ub_law.jpg | tail -n 2
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565  ..RT.8..o..Z.T.e
0019d1f0: d351 9ebf ffd9                .Q....

(kali@kali)-[~/datacarving_lab3]
$
```

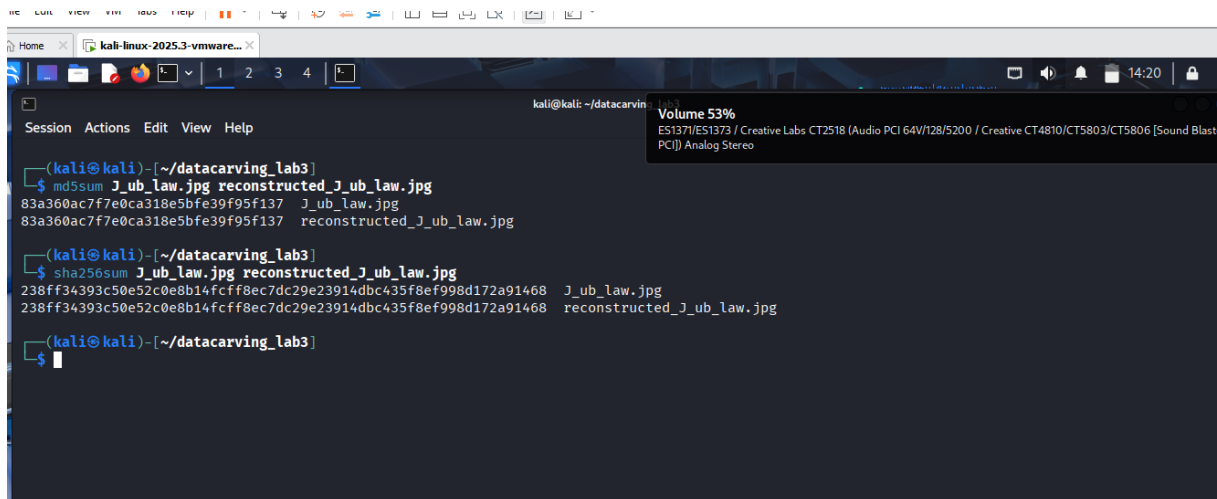


A terminal window titled 'kali@kali: ~/datacarving_lab3' showing the conversion of the image file to a hex file and then back to a reconstructed image. The first command is 'xxd -p J_ub_law.jpg > J_ub_law.hex'. The second command is 'xxd -r -p J_ub_law.hex reconstructed_J_ub_law.jpg'. A system notification in the top right corner indicates 'Computer 56% - 3 hours 29 minutes remaining'.

```
(kali@kali)-[~/datacarving_lab3]
$ xxd -p J_ub_law.jpg > J_ub_law.hex

(kali@kali)-[~/datacarving_lab3]
$ xxd -r -p J_ub_law.hex reconstructed_J_ub_law.jpg

(kali@kali)-[~/datacarving_lab3]
$
```



```
kali@kali: ~/datacarving_lab3
$ md5sum J_ub_law.jpg reconstructed_J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137 J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137 reconstructed_J_ub_law.jpg

(kali@kali)~/datacarving_lab3
$ sha256sum J_ub_law.jpg reconstructed_J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 J_ub_law.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 reconstructed_J_ub_law.jpg

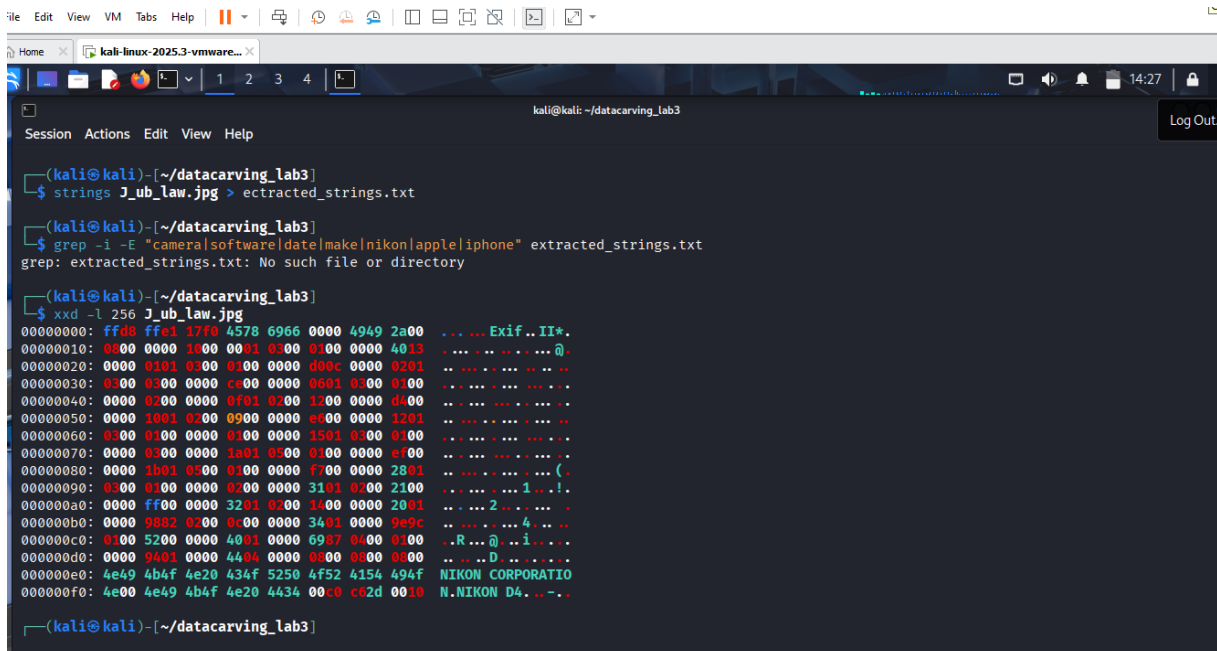
(kali@kali)~/datacarving_lab3
$
```

Part B - Examine Embedded Metadata-Like Content

Use xxd and strings to inspect the JPEG for visible camera, software, name, date or other metadata-related strings.

Record only information actually found. If a searched item is not present, state “**not found**” rather than guessing.

ANSWER:



```
kali@kali: ~/datacarving_lab3
$ strings J_ub_law.jpg > extracted_strings.txt

(kali@kali)~/datacarving_lab3
$ grep -i -E "camera|software|date|make|nikon|apple|iphone" extracted_strings.txt
grep: extracted_strings.txt: No such file or directory

(kali@kali)~/datacarving_lab3
$ xxd -l 256 J_ub_law.jpg
00000000: ffd8 ffe1 17f9 4578 6966 0000 4949 2a00  ... Exif..II*.
00000010: 0000 0000 1000 0001 0300 0300 0000 4013  ... ..
00000020: 0000 0101 0300 0300 0000 d00c 0000 0201  ... ..
00000030: 0300 0300 0000 c000 0000 0001 0300 0100  ... ..
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400  ... ..
00000050: 0000 1001 0200 0900 0000 e600 0000 1201  ... ..
00000060: 0300 0300 0000 0100 0000 1501 0300 0100  ... ..
00000070: 0000 0300 0000 1a01 0300 0100 0000 ef00  ... ..
00000080: 0000 1b01 0300 0100 0000 f700 0000 2801  ... ..
00000090: 0300 0300 0000 0200 0000 3101 0300 2100  ... ..
000000a0: 0000 ff00 0000 3201 0200 1400 0000 2001  ... ..
000000b0: 0000 9882 0200 0600 0000 3401 0000 9e9c  ... ..
000000c0: 0100 5200 0000 4001 0000 6987 0400 0100  ... R...@...i...
000000d0: 0000 9401 0000 4404 0000 0800 0800 0800  ... ..D...
000000e0: 4e49 4b4f 4e20 434f 5250 4f52 4154 494f  NIKON CORPORATIO
000000f0: 4e00 4e49 4b4f 4e20 4434 00c0 c02d 0010  N.NIKON D4...
(kali@kali)~/datacarving_lab3
```

Part C - Direct Data-Unit Examination with The Sleuth Kit

Use tools such as:

img_stat
mmls
fsstat
fls
icat
blkcat

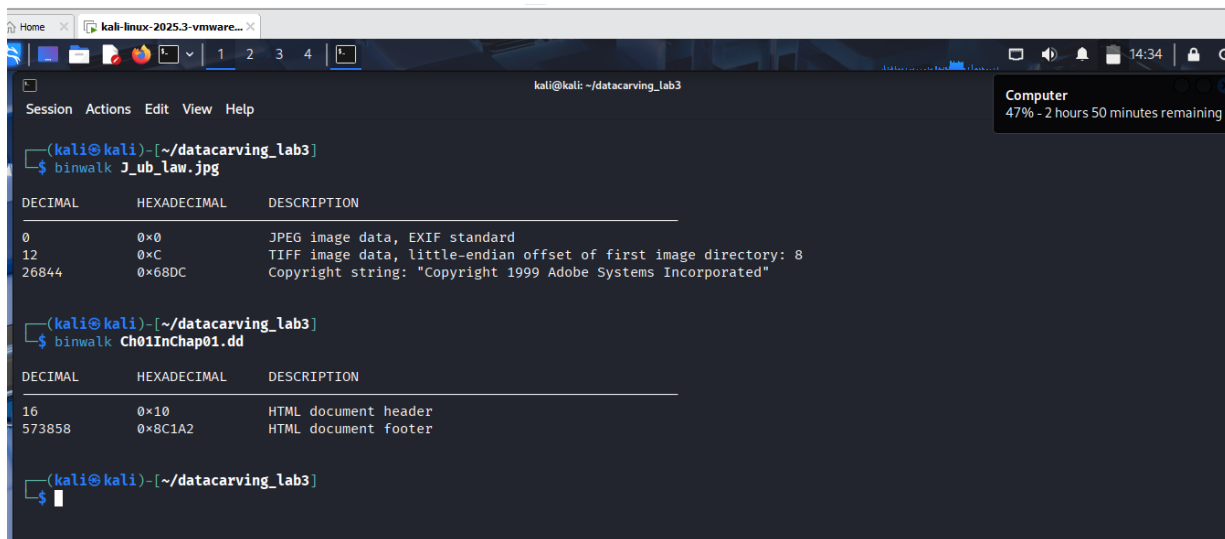
Identify actual inode and sector/block values from your own command output. Do not copy placeholder values from examples.

Part D - Embedded File Discovery with Binwalk

Use binwalk to identify embedded file structures or signatures.

If the instructor-provided File_carving.docx is available, perform the demonstrated manual and automatic extraction workflow. If it is not available, document this limitation and complete the Binwalk practice using the other supplied lab evidence.

ANSWER:



```
kali@kali: ~/datacarving_lab3
$ binwalk J_ub_law.jpg
DECIMAL      HEXADECI-  DESCRIPTION
MAL          MAL
0            0x0        JPEG image data, EXIF standard
12           0xC        TIFF image data, little-endian offset of first image directory: 8
26844       0x68DC     Copyright string: "Copyright 1999 Adobe Systems Incorporated"

(kali@kali)~/datacarving_lab3
$ binwalk Ch01InChap01.dd
DECIMAL      HEXADECI-  DESCRIPTION
MAL          MAL
16           0x10       HTML document header
573858       0x8C1A2    HTML document footer

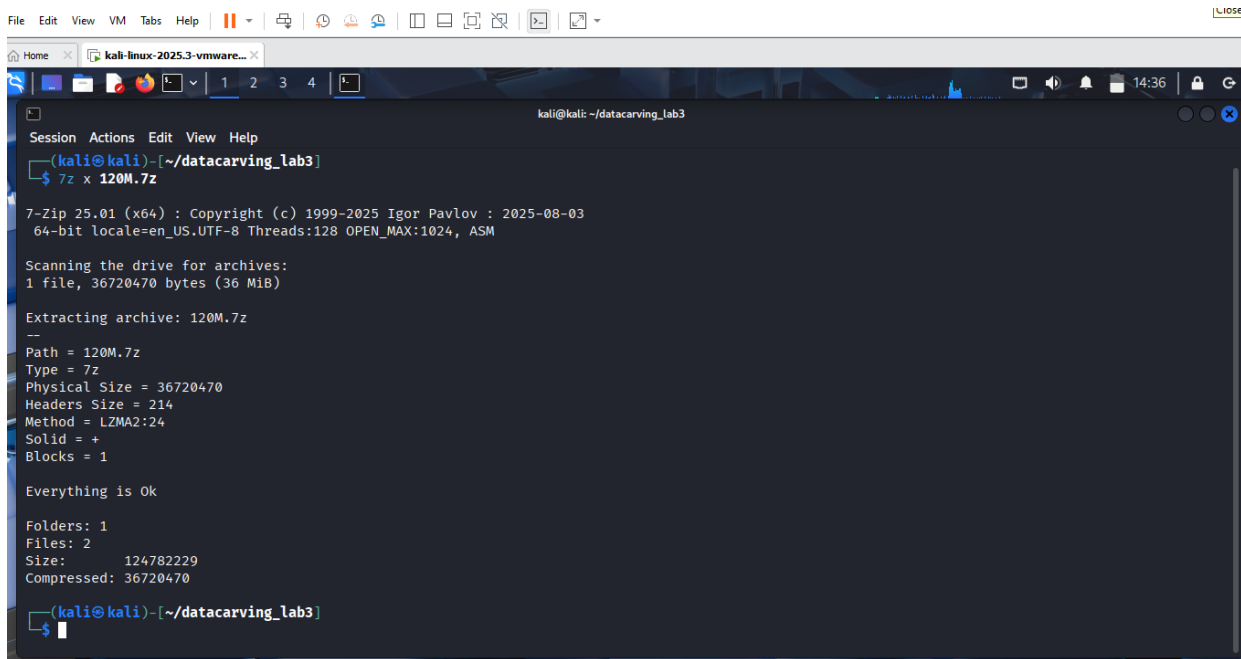
(kali@kali)~/datacarving_lab3
$
```

Part E - Prepare the USB Image

Extract 120M.7z, identify the resulting USB image, calculate hashes, inspect the image using TSK, and confirm the appropriate partition offset using mmls.

The source lab refers to an offset of 128, but students must confirm the partition structure rather than blindly copying the value.

ANSWER:



```
kali@kali: ~/datacarving_lab3
Session Actions Edit View Help
(kali@kali)-[~/datacarving_lab3]
$ 7z x 120M.7z

7-Zip 25.01 (x64) : Copyright (c) 1999-2025 Igor Pavlov : 2025-08-03
64-bit locale=en_US.UTF-8 Threads:128 OPEN_MAX:1024, ASM

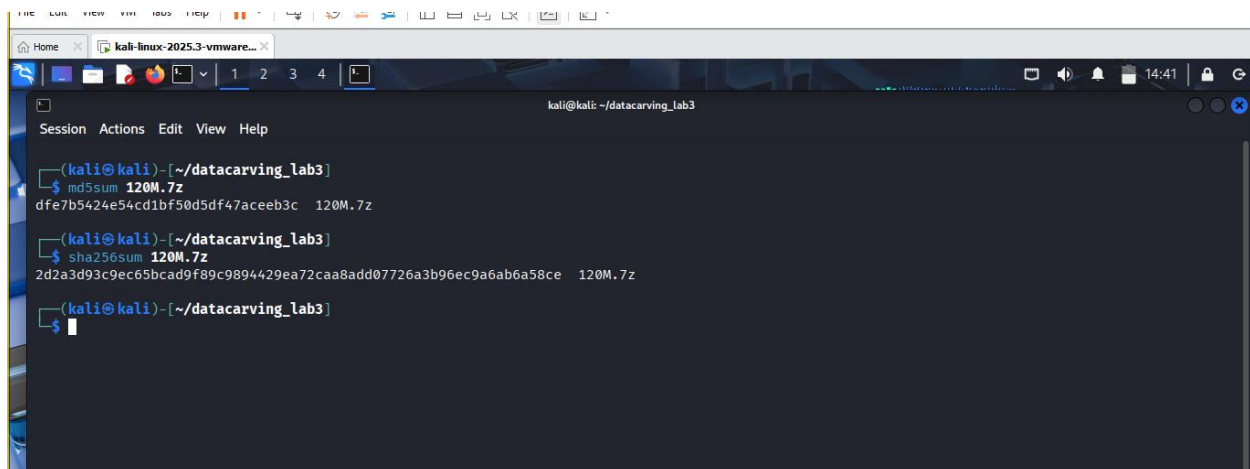
Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Extracting archive: 120M.7z
--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

Everything is Ok

Folders: 1
Files: 2
Size:      124782229
Compressed: 36720470

(kali@kali)-[~/datacarving_lab3]
$
```



```
kali@kali: ~/datacarving_lab3
Session Actions Edit View Help
(kali@kali)-[~/datacarving_lab3]
$ md5sum 120M.7z
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z

(kali@kali)-[~/datacarving_lab3]
$ sha256sum 120M.7z
2d2a3d93c9ec65bcd9f89c9894429ea72caa8add07726a3b96ec9a6ab6a58ce 120M.7z

(kali@kali)-[~/datacarving_lab3]
$
```

Part F - Carve Deleted Files with Scalpel

Configure Scalpel to enable the required file types, particularly JPEG/JPG, and run the carving operation against the forensic USB image.

Review:

- Recovered files

- File types
- Hashes
- audit.txt
- At least two recovered JPEG images where available

```

kali@kali: ~/datacarving_lab3
$ sudo rm -r carved_output_new

kali@kali: ~/datacarving_lab3
$ sudo scalpel -c /etc/scalpel/scalpel.conf -o carved_output_new/ Ch01InChap01.dd
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/kali/datacarving_lab3/Ch01InChap01.dd"

Image file pass 1/2.
Ch01InChap01.dd: 100.0% |*****| 560.4 KB 00:00 ETA
Allocating work queues...
Work queues allocation complete. Building carve lists...
Carve lists built. Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x45\x78\x69\x66" and footer "\xff\xd9" -> 0 files
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" -> 0 files
Carving files from image.
Image file pass 2/2.
Ch01InChap01.dd: 100.0% |*****| 560.4 KB 00:00 ETA
Processing of image file complete. Cleaning up...
Done.
Scalpel is done, files carved = 0, elapsed = 0 seconds.

kali@kali: ~/datacarving_lab3
$

```

Document every command, result, screenshot, recovered artefact, error and limitation.

Do not modify the original evidence files.

Required evidence

Submit one professional PDF forensic report and one ZIP archive containing the required recovered/carved evidence files, hash records and supporting output.

Required evidence must include:

Proof that the Module 7 material was reviewed before starting.

Working directory and downloaded evidence files with file sizes and hashes.

xxd output showing JPEG header and footer analysis.

Reverse-hexdump reconstruction and hash comparison.

strings or metadata-style examination of the sample JPEG.

TSK mmls / fls and related data-unit analysis output.

Scalpel configuration, execution output and audit.txt.

At least two recovered JPEG files opened successfully, or clear evidence explaining why they could not be opened.

Lab access notes This practical forms part of the second laboratory assessment window. Students must review the Module 7 lessons and study material before beginning. Use only the authorised training files provided in the lab instructions. Preserve the original evidence, record hashes, document all commands and submit both the required PDF report and supporting evidence archive.